Online Traffic Fine Payment System (JURMANA)

Bachelor of Technology
Computer Science and Engineering

Submitted By

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1. Introduction

The **Jurmana** project aims to design a digital system that automates the process of fine collection and management within an institution or organization. It provides a transparent and efficient way to record, track, and manage fines imposed on students or employees for various violations. The system ensures proper communication, real-time updates, and easy payment management.

The software will be developed as a **web-based application** with features like fine generation, payment tracking, reports, and notifications. It will allow admin-level users to create, update, or delete fines and users to check and pay their dues. The project will be developed using **Java, MySQL, and modern web technologies**, ensuring scalability, reliability, and user-friendliness.

I. Project Overview

The **Jurmana System** will automate manual fine management activities in academic and corporate environments. It will reduce human error and improve accountability by recording all transactions digitally.
Users will be able to:

- View and pay fines online.
- Receive notifications for unpaid fines.
- Download payment receipts.
- Admins can track fine history, generate reports, and manage users.

The goal is to create a secure, easy-to-use system that saves time and ensures accurate recordkeeping.

II. Project Deliverables

1. Preliminary Project Plan	05.06.2025
2. Requirements Specification	15.06.2025
3. Analysis [Object model, Dynamic model, and User interface]	22.06.2025
4. Architecture Specification	30.06.2025
5. Component/Object Specification	10.06.2016
6. Source Code	20.07.2025- 30.07.2025
7. Test Plan	31.07.2025 - 07.08.2025
8. Final Product Demo	10.08.2025 - 15.08.2025

III. Evolution of this Document

This document will evolve as the project progresses. Updates may occur in:

- **References** – as new materials are reviewed.
- **Organizational Structure** – as new team roles are assigned.
- **Technical Process** – as requirements and design details are refined.
  Schedule – adjusted per project progress.

REVISION HISTORY

Revision	Date	Updated By	Update Comments
0.1	05.062025	Ragyam Sinha	First Draft
0.2	15.06.2025	Ragyam Sinha	Second Draft/Final Draft

IV. References

- Pressman, Roger S. Software Engineering: A Practitioner's Approach.
- Sommerville, Ian. Software Engineering (10th Edition).
- Martin, Robert C. Clean Code: A Handbook of Agile Software Craftsmanship.
- Payroll Management System – Concepts & Implementation, Tutorialspoint. https://www.tutorialspoint.com/payroll-management-system
- Understanding Payroll in India, Razorpay Payroll Documentation. https://razorpay.com/payroll
- Payroll Software Features & Compliance Guide, Zoho Payroll. https://www.zoho.com/payroll
- Designing Web Applications Using PHP & MySQL, W3Schools. https://www.w3schools.com/php
- Best Practices for Database Design and Normalization, MySQL Documentation. https://dev.mysql.com/doc
- Introduction to Software Testing, Guru99. https://www.guru99.com/software-testing.html

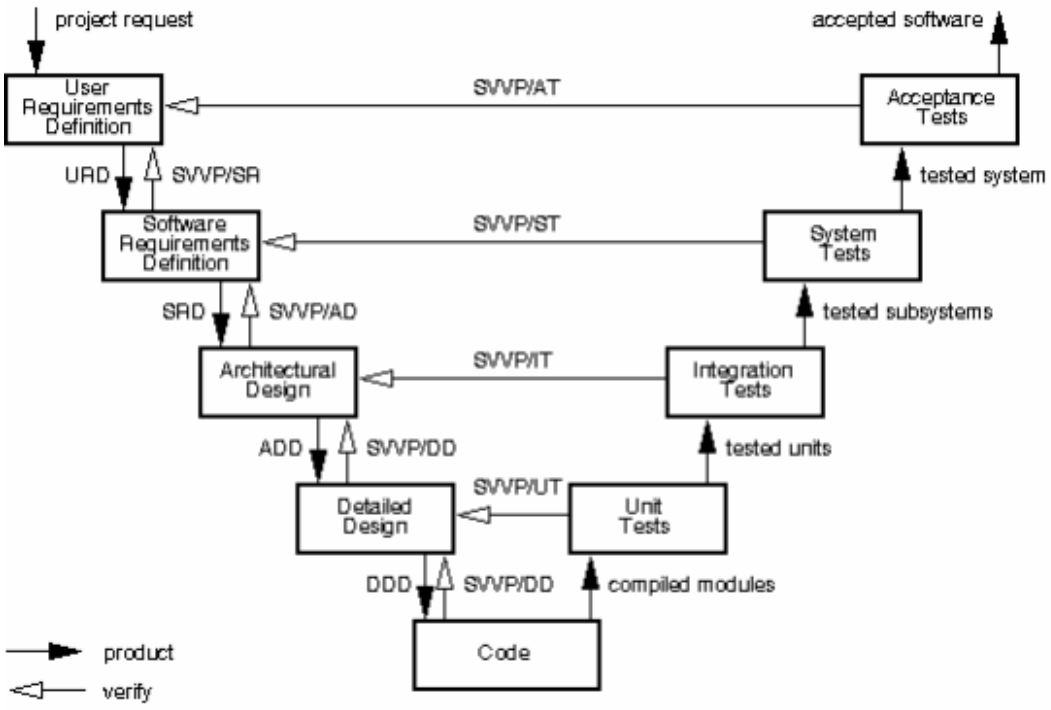
V. Deffinitions, Acronyms, and Abbrevations

UML – Unified Modeling Language
AD – Architectural Design
ADD – Architectural Design Document
DD – Detailed Design
DDD – Detailed Design Document
PM – Project Manager
QAM – Quality Assurance Manager
SM – Senior Management
SPMP – Software Project Management Plan (this document)
SRS – Software Requirements Specification
OPMS – Online Payroll Management System
UI – User Interface
PF – Provident Fund
ESI – Employee State Insurance
TDS – Tax Deducted at Source
RBAC – Role-Based Access Control
CRUD – Create, Read, Update, Delete (basic operations used in the system)

2. Project Organization

I. Process Model

The **V-Model** process is used, where each development stage is followed by corresponding testing. This ensures that design and implementation are continuously validated.


project request
accepted software
User Requirements Definition
SVVP/AT
Acceptance Tests
URD
SVVP/SR
tested system
Software Requirements Definition
SVVP/ST
System Tests
SRD
SVVP/AD
tested subsystems
Architectural Design
SVVP/IT
Integration Tests
ADD
SVVP/DD
tested units
SVVP/UT
Detailed Design
Unit Tests
DDD
SVVP/DD
compiled modules
Code
product
verify

II. Organization Structure

Team Members –

- Ragyam Sinha – Project Manager
- Aditya Sharma – Analyst
- Priya Das – Designer
- Ritu Kumar – Developer
- Arjun Patel – Tester

Name	Organization/ Position	Contact Information
Ragyam Sinha	Project Manager	ragyamsinha@tmsl.com
Aditya Sharma	Analyst	adityasharma@tmsl.com
Priya Das	Designer	priyadas@tmsl.com
Ritu Kumar	Developer	Ritu.kumar@tmsl.com
Arjun Patel	Tester	Arjun.patel@tmsl.com

Days	Deliverable	Team Leader	Deliverable Description
9	1	Ragyam Sinha	Project Plan
7	2	Sumit Kumar	Requirements Specification
9	3	Utkarsh Kumar Jaiswal	Analysis
13	4	Surbhi Kumari	Architecture Specification
9	5	Vaisnavi	Component/Object Specification
14	6	Ravi Kumar Gupta	Source Code
7	7	Shubham Singh	Test Plan
5	8	Arnab Mondal	Final Deliverable

III. Organizational Boundaries and Interfaces

Each team leader is responsible for managing deliverables, internal communication, and progress tracking. Collaboration tools such as Google Meet and GitHub will be used for coordination.

IV. Project Responsibilities

Deliverable	Team Leader	Description
Project Plan	Ragyam Sinha	Define scope, schedule, and milestones
Requirements Specification	Aditya Sharma	Define system functionalities
Analysis	Priya Das	Create UML diagrams and flowcharts
Architecture Specification	Ritu Kumar	Define software structure
Source Code	Ritu Kumar	Implement backend and frontend
Test Plan	Arjun Patel	Plan and execute test cases
Final Deliverable	Entire Team	Integration and presentation

3. Managerial Process

I. Management Objective and Priorities

Runner will be utilized. The risks for each category are listed to complete the project successfully. For each risk, a description, a probability of occurrence, the associated action and the impact of the risk are given.

- Project Charter
- SRS
- Design Documents (UML, DFD, ER Diagram)
- Test Plan and Reports
- Implementation Plan
- Final Project Report

II. Assumptions, Dependencies and Constrains

- Team of 5 members × 300 hours = 1500 hours total.
- Deadline: **15th August 2025**.
- All coding done in **Java with MySQL backend**.
- Weekly status meetings and internal deadlines.
- Limited hardware and no external funding.

III. Risk Management

This section mentions any potential risks for the project. Also, schedules or methods are defined to prevent or to reduce the risks as below:

i. Technology risk
ii. People risk
iii. Financial risk
iv. Market risk
v. Structure/process risk

The following are the possible risks to be encountered during the development of the project and how they can be prevented.

1. Miscommunication

Prevention: Team members should not hesitate to ask and re-ask questions if things are unclear. Team members should have a written copy of the tasks assigned to them every meeting.

Correction: When it becomes clear that miscommunication is causing problems, the team members should gather in a meeting to clear things up.

2. Time shortage

Prevention: Care is taken to plan enough spare time.

Correction: When tasks fail to be finished in time or when they are finished earlier than planned the project planning is adjusted

3. Illness or absence of team members

Prevention: Team members should warn their team leader or the PM timely before a planned period of absence.

Correction: Work can be taken over quickly by someone else or be distributed among the team members if a person gets ill.

4. Monitoring and Controlling Mechanisms:

The monitoring of progress is done by the PM using the following means:

Project Kick-off Meetings -The project group meetings take place within the class room or through chat. These meetings are meant to inform each other of the progress made on various tasks and to assign new tasks.

Progress Report- Progress report is done every Friday. This is meant to inform and show the progress in the development of the project and how things are going.

Risk	Prevention	Correction
Miscommunication	Weekly team syncs	Clarify tasks in writing
Time Shortage	Strict scheduling	Adjust priorities
Technical Issues	Early testing	Backup versions
Absenteeism	Task rotation	Redistribute workload

IV. Monitoring and Controlling Mechanisms

The monitoring of progress is done by the PM using the following means:

- Weekly project status meetings
- Shared GitHub repository
- Progress tracked using MS Project
- Document revisions stored in shared drive

4. Technical Process

I. Methods, Tools, and Techniques

The development of the **Online Traffic Fine Payment System (OTFPS – JURMANA)** will follow the **Iterative Waterfall Model**, ensuring a structured and systematic approach across all phases:

- **Requirements Analysis**
- **System and Architectural Design**
- **Development and Integration**
- **Testing and Verification**
- **Deployment and Review**

Each iteration will refine and expand system features such as:
user authentication, **fine management**, **online payments**, **dispute handling**, and **notification services**.

Tools and Technologies Used

The project will be built using the following tools and technologies:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js / Express *(or PHP if chosen)*
- **Database:** MySQL
- **Development Environment:** Visual Studio Code
- **Modeling and Diagrams:** StarUML (DFD, UML, ER diagrams)
- **Testing Tools:** Browser DevTools, Postman (API testing), MySQL Workbench

Testing Approach

Testing will be conducted using:

- **Manual Functional Testing** (fine details, payment flow, dispute submission)
- **UI and Browser Compatibility Testing**
- **API Testing** (payment, login, fine retrieval)
- **Database Validation** (record accuracy, updates, and transactions)

Risk Identification & Mitigation

Common risks related to technology, tools, and methodology will be documented with:

- **Risk Description**
- **Probability (Low/Medium/High)**
- **Impact Level**
- **Mitigation Strategy**

Examples include payment gateway failures, API downtime, user authentication errors, and inaccurate location detection.

This risk management ensures smooth project execution and successful completion.

II. Software Documentation

Documentation for the **Online Traffic Fine Payment System (OTFPS – JURMANA)** will include all necessary technical and user-oriented documents to support development, testing, deployment, and maintenance. These include:

- **Software Requirements Specification (SRS)**
- **System Analysis and Design Document**
- **Data Flow Diagrams and UML Diagrams**
- **Technical Specification Document**
- **Database Schema & ER Diagram**
- **Detailed Design Document**
- **Test Plan and Test Case Document**
- **Implementation Plan**
- **User Manual / User Guide**
- **Final Project Report**

These documents will ensure clarity for developers, testers, administrators, and end users, supporting long-term maintainability and scalability of the system.

5. Work Elements, Schedule and Budget

a) Work Elements

The project includes all necessary tasks, tools, and technologies required to complete the analysis, design, development, testing, and deployment of the OTFPS – JURMANA system.

b) Team Responsibilities

Project responsibilities will be shared and rotated among the three team members, with each member leading specific phases to ensure equal contribution and balanced workload throughout the project.

c) Documentation

All documents produced in every phase—such as requirements, design, development, and testing—will be reviewed regularly and updated as needed to reflect any changes made during the project lifecycle.

Category	Amount (INR)
Salary / Team Effort Value	2,20,000.00
Operations / Supplies / Tools / Equipment	1,55,000.00
Miscellaneous Expenses	9,392.6368
Total Estimated Budget	**₹3,84,392.6368**

	Name	Duration	Start	Finish
1	**Project Initialization**	**11 days?**	**12/11/25, 8:00am**	**26/11/25, 5:00pm**
2	Feasibility Study	4 days?	12/11/25, 8:00am	17/11/25, 5:00pm
3	Requirement Gathering	3 days?	18/11/25, 8:00am	20/11/25, 5:00pm
4	Requirement Analysis	4 days?	21/11/25, 8:00am	26/11/25, 5:00pm
5	**Design**	**12 days?**	**27/11/25, 8:00am**	**12/12/25, 5:00pm**
6	System Design	4 days?	27/11/25, 8:00am	02/12/25, 5:00pm
7	Database Design	4 days?	03/12/25, 8:00am	08/12/25, 5:00pm
8	GUI Design	4 days?	09/12/25, 8:00am	12/12/25, 5:00pm
9	**Coding**	**16 days?**	**15/12/25, 8:00am**	**05/01/26, 5:00pm**
10	Frontend	5 days?	15/12/25, 8:00am	19/12/25, 5:00pm
11	Backend	8 days?	22/12/25, 8:00am	31/12/25, 5:00pm
12	Database Model	3 days?	01/01/26, 8:00am	05/01/26, 5:00pm
13	**Testing**	**9 days?**	**06/01/26, 8:00am**	**16/01/26, 5:00pm**
14	Unit Testing	3 days?	06/01/26, 8:00am	08/01/26, 5:00pm
15	Integration Testing	3 days?	09/01/26, 8:00am	13/01/26, 5:00pm
16	System Testing	3 days?	14/01/26, 8:00am	16/01/26, 5:00pm